

Stochastic-cascade parallel track: Hasegawa–Wakatani bridge

Living project note

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1 Purpose

This note records the exploratory stochastic-cascade parallel track to the canonical finite-dimensional nonmodal theory. The proposed chain is

field dynamics $\rightarrow$ zonal/nonzonal split $\rightarrow$ scale process $\rightarrow$ Markov test $\rightarrow$ Kramers–Moyal diagnostics.

It does not choose the final plasma convention for the main project and it does not constitute a converged turbulence calculation.

2 Benchmark equations

The modified Hasegawa–Wakatani (mHW) pilot uses

$$\partial_t \zeta + \{\phi, \zeta\} = \alpha(\tilde{\phi} - \tilde{n}) - D_\zeta \nabla^4 \zeta, \quad (1)$$

$$\partial_t n + \{\phi, n\} = \alpha(\tilde{\phi} - \tilde{n}) - \kappa \partial_y \phi - D_n \nabla^4 n, \quad (2)$$

with

$$\zeta = \nabla_\perp^2 \phi, \quad \tilde{f} = f - \langle f \rangle_y.$$

The Numata-style modification removes the resistive/adiabatic coupling from zonal modes $k_y = 0$. C2 also uses an original-HW control in which the same coupling is retained on zonal modes.

All C2 runs use a 32^2 periodic grid with $L_x = L_y = 40$, $\alpha = \kappa = 1$, $D_\zeta = D_n = 0.02$, a pseudo-spectral Poisson bracket, two-thirds dealiasing, RK4 with $\Delta t = 0.05$, and random initial perturbations of amplitude 10^{-3} .

3 C1: single-realization bridge pilot

The first pilot used seed 3 and the transient window $120 \leq t \leq 310$. It established that the full diagnostic chain can be implemented and gave an initial low/high-zonal-flow split. For the scale step $4\Delta x \rightarrow 2\Delta x$, fits of

$$D^{(1)}(u) \approx a_1 u + a_3 u^3, \quad D^{(2)}(u) \approx b_0 + b_2 u^2$$

gave

coefficient	low ZF	high ZF
a_1	-0.72497	-0.72289
a_3	0.00337	0.00207
b_0	0.34951	0.11664
b_2	0.19309	0.17742

The factor-of-three drop of b_0 was treated only as a hypothesis because ZF strength was strongly confounded with time.

4 C2: independent-seed falsification

C2 integrates four independent seeds, 0, 1, 2, 3, from $t = 0$ to 500 and analyzes only $380 \leq t \leq 490$. The mean zonal kinetic-energy fractions are

seed	mHW	HW
0	0.93080	0.05864
1	0.95111	0.04578
2	0.95755	0.04193
3	0.90398	0.04823

The mHW ZF fraction is near a late plateau, although the residual nonzonal energy still has seed-dependent slow trends. The window is therefore called late or quasi-stationary, not rigorously stationary.

4.1 Direction-resolved diffusion signal

Each mHW seed is split at its own median ZF fraction. Radial (x) and poloidal (y) increments are analyzed separately. Seed-level bootstrap intervals use 20,000 resamples of the four independent realization-level estimates.

For the $4\Delta x \rightarrow 2\Delta x$ step,

direction	mean $b_0^{\text{high}}/b_0^{\text{low}}$	95% interval
x	0.80188	[0.71009, 0.89366]
y	0.86447	[0.84857, 0.88448]

All four seeds have $b_0^{\text{high}} < b_0^{\text{low}}$ in both directions. The original C1 magnitude was therefore too large, but the sign of the diffusion modulation survives the independent-seed and anisotropy controls.

The corresponding radial a_1 change is consistent with zero; hence the dominant reproducible C2 signal remains diffusion-like rather than drift-like.

4.2 Markov-improvement hypothesis is falsified

A discrete Chapman–Kolmogorov test compares

$$P(X_2 | X_0) \quad \text{with} \quad P(X_2 | X_1)P(X_1 | X_0).$$

For the 8–4–2 triplet, the high-minus-low ZF change in occupancy-weighted total-variation error is

$$\Delta_{\text{CK}}^x = -0.00306 \quad [-0.00745, 0.00452],$$

while

$$\Delta_{\text{CK}}^y = +0.01341 \quad [0.00366, 0.02119].$$

Thus stronger ZF does not generally improve Markov closure. The radial change is unresolved at this seed count and the poloidal error becomes worse. Markov closure must remain scale- and direction-dependent and is an independent gate.

4.3 Original-HW control

Each model, seed and direction is normalized by its own $r = 8\Delta x$ standard deviation so that the comparison is not merely a comparison of raw fluctuation amplitudes. Even after this self-normalization, mHW has lower finite-step diffusion intercept than HW:

step	x : mHW/HW	95% interval	y : mHW/HW	95% interval
$8 \rightarrow 7$	0.70686	[0.67239, 0.73490]	0.52698	[0.48195, 0.57200]
$8 \rightarrow 4$	0.73844	[0.64518, 0.83361]	0.61841	[0.57258, 0.68577]

The conservative interpretation is therefore

strong zonal organization is associated with reduced finite-step stochastic spreading of the nonzonal scale pro

This remains an association statement: changing from HW to mHW changes the equations, not only the instantaneous ZF amplitude.

5 Small-step and Pawula gate

At fixed current separation $r = 8\Delta x$, C2 evaluates next separations

$$r' = 7, 6, 5, 4 \Delta x, \quad \Delta s = \ln(8/r').$$

The smallest available scale step is $\ln(8/7) \simeq 0.1335$.

The mean fitted b_0 values do not form a scale-step-independent plateau:

case	$8 \rightarrow 7$	$8 \rightarrow 6$	$8 \rightarrow 5$	$8 \rightarrow 4$
mHW x	0.52627	0.66145	0.55804	0.41018
mHW y	0.51473	0.65760	0.61463	0.45771
HW x	0.74600	0.92691	0.77099	0.55781
HW y	0.98276	1.21100	1.01931	0.74598

For the finite-step fourth coefficient

$$D_{\Delta s}^{(4)}(u) = \frac{\langle (\Delta u)^4 | u \rangle}{24 \Delta s},$$

a simple linear extrapolation of its constant fit coefficient gives nonzero $\Delta s \rightarrow 0$ intercepts:

case	extrapolated c_0	95% interval
mHW x	0.03186	[0.02945, 0.03426]
mHW y	0.02196	[0.01266, 0.03125]
HW x	0.05968	[0.05430, 0.06444]
HW y	0.09816	[0.08597, 0.11034]

The extrapolation itself is only a resolution diagnostic. The appropriate conclusion is that the present 32^2 data do not justify a Pawula/Fokker–Planck truncation.

6 Current gate decision

- **Passed:** a ZF-conditioned reduction of the finite-step second conditional moment survives independent seeds, anisotropy separation, and an original-HW control after self-normalization.
- **Falsified:** a general claim that stronger ZF improves Markov closure.
- **Failed at current resolution:** convergence of $D^{(1)}$, $D^{(2)}$ as $\Delta s \rightarrow 0$ and a Pawula justification for truncation at second order.

The robust object at this stage is therefore the *ZF-conditioned finite-scale transition kernel*, not yet a Fokker–Planck cascade.

Before transfer to GENE, the next gate is higher spatial resolution together with alternative filtered/wavelet scale variables. Only if $D^{(2)}$ develops a small-step plateau and $D^{(4)}$ moves toward zero should the finite-step quantities be promoted to genuine Kramers–Moyal coefficients.

7 References

R. Numata, R. Ball, and R. L. Dewar, *Nonlinear Simulation of Drift Wave Turbulence*, arXiv:physics/0703274 (2007).

A. Hakim, *Solving (Modified) Hasegawa–Wakatani equations*, Simulation Journal JE17 (2026).